Conditional generation of cathode vacancy patterns at requested compositions and energy ranges

Ricardo Silva Buarque, Beihao Gu, and Varad Kottawar
ECE 285 midterm draft — November 9, 2026

Abstract

We extend an existing energy-conditioned autoregressive (AR) conditional variational autoencoder (CVAE) to request low, middle, or high energy Li/vacancy patterns in one fixed layered $LiCoO_2$ (O3) parent. A separate acquisition policy allocates requests using static machine-learned interatomic-potential (MLIP) energies at relevant operating temperatures. Under UMA, novel in-bin yield is 31.0% for multibin, 33.0% for uniform sampling and 58.4% for bin-seeded mutation: the extension does not improve control. These are teacher-defined generation and acquisition tests, not measurements of physical configurational entropy or reactor equilibrium.

1 Physical motivation and revised objective

Direct cathode recycling seeks to recover useful composition and structure. Hydrothermal treatment and subsequent annealing have regenerated degraded $LiCoO_2$ (LCO) [4]. Our longer-term application is to select informative reference structures for better models of lithium replenishment. The present class problem is controlled generation, rather than a simulation of the recycling liquid or reaction pathway.

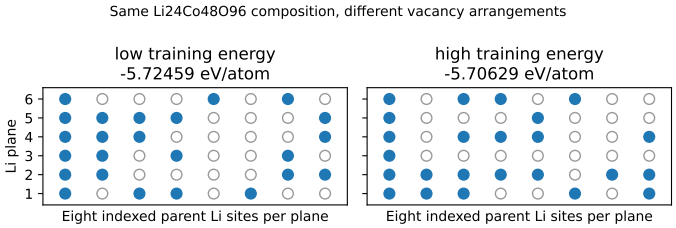
Density functional theory (DFT) computes an electronic-structure reference at specified atoms and cell under a stated approximation and numerical protocol. An MLIP approximates the energy and its force/cell derivatives cheaply. Molecular dynamics (MD) propagates atoms with those forces; appropriately sampled trajectories can estimate thermal observables. A single-point call fixes coordinates; a relaxation moves them toward a local minimum. Drawing a vacancy mask does neither operation.

Li/vacancy ordering can change energies at a fixed lithium content, as established in first-principles LCO studies [6]. Foundation datasets have broad equilibrium and displaced-structure coverage, but do not guarantee the needed defect/order coverage; OMat24 discusses these limitations explicitly [2]. We do not assume all teacher errors are constant offsets or that every foundation model was trained only on pristine LCO.

2 Parent, geometry protocol, and symmetry

The main parent is $Li_{48}Co_{48}O_{96}$. Retaining n Li gives $N_a = 144 + n$ present atoms and dimensionless Li fraction $x = n/48$. Occupancy $o_j = 1$ means parent Li site j is occupied; vacancy $v_j = 1 - o_j$ means removed. Li are full-parent atom rows 48–95 (zero based), not the first rows. The controlled comparison uses $n = 12, 24, 36$. Labels refer to unchanged, pristine-derived deletion coordinates, not relaxed minima.

The parent's 192 retained site permutations identify equivalent masks. Their orbit is the set of distinct related raw occupation patterns; its canonical key is the smallest transformed vacancy string. Grouping precedes splitting and augmentation; all frames/teacher labels of a lineage stay together. At $n = 24$, Burnside counting gives 168,025,313,938 orbits for this group, compared with 32,247,603,683,100 raw masks. This is a count for the specified cell, not all 48-site cell shapes or host phases. O1/spinel transitions and size/material transfer are outside this fixed-parent generator.



Two existing Li_{24} training deletion patterns: filled dots are occupied indexed sites in each Li plane. The columns are site labels, not an x–y projection. Teacher energies are in eV/atom; these are neither new CVAE discoveries nor relaxed structures. POSCAR/CIF and source hashes are supplied.

3 Composition-and-energy-range CVAE

The model already used composition and a binary low-energy class. The new schema extends this condition to three energy ranges. At each composition, define $e_i = E_i/N_a$ in eV/atom, where E_i is total cell energy in eV. Sort the m *training* energies; the one-based element at $\lceil pm \rceil$ defines q_p. The requested bin b is

$$b(e) = \begin{cases} \text{low}, & e \le q_{0.2}, \\ \text{middle}, & q_{0.2} < e \le q_{0.8}, \\ \text{high}, & e > q_{0.8}. \end{cases} \tag{1}$$

Exact ties stay together. Empty bins are explicitly unsupported; sparse compositions are not forced to have three classes. The versioned schema stores edges, support, units, teacher/head, geometry protocol, parent/data/split hashes, and content identity. It is shared by fitting, generation, allocation and assessment. It uses no validation or assessment energies to fit the boundaries.

The condition is $c = [x, \text{onehot}(b)]$, with four components. Historical binary $[x, \mathbf{1}_{\text{low}}]$ and composition-only $[x]$ controls remain functional with their original input dimensions. Binary middle/high requests explicitly map to its rest class. Incompatible checkpoint shapes are rejected; energy conditioning itself is not a new invention here.

The encoder $q_\phi(z \mid o, c)$ predicts the mean and diagonal variance of an eight-dimensional Gaussian. At generation, $z \sim \mathcal{N}(0, I)$ and c enter a gated recurrent unit (GRU) decoder with hidden size 64. Its normalized autoregressive likelihood is

$$p_\theta(o \mid z, c) = \prod_{j=1}^{48} p_\theta(o_j \mid o_{<j}, z, c). \tag{2}$$

Site index j is a sequence position, not MD time. Observed preceding bits are used during fitting (teacher forcing); sampled bits are fed back at generation. When the remaining Li count forces a zero or one, that choice has probability one and zero reconstruction penalty. Thus every valid sample has exactly n Li. The chain-rule construction is related to variational autoregressive approaches [7], but our supervised conditional objective differs.

The loss is

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}[-\log p_\theta(o \mid z, c)] + \beta D_{\text{KL}}[q_\phi(z \mid o, c) \| \mathcal{N}(0, I)]\,. \tag{3}$$

Parameters ϕ, θ belong to encoder and decoder; I is the identity matrix, and $\mathbb{E}$ denotes expectation. KL measures the encoded Gaussian's difference from the prior, with weight β. At $\beta = 1$, this is a negative conditional evidence lower bound; $\beta = 0.1$ is a different regularized objective [3, 5]. Good reconstruction or large KL alone does not prove good prior samples [1]. Requesting low energy does not impose an energy inequality or provide a minimization gradient.

3.1 Data, controls, and assessment

The primary expanded round-0 set contains 50,247 orbit masks: 40,157 training, 4,967 validation and 5,123 test rows in protected splits. New fit configurations record actual seeds, update/stopping limits and checkpoint selection. Previously assessed data promoted to training become development evidence. A fresh assessment boundary and prior-history cache are frozen before new proposals.

Controls are binary/composition CVAEs, uniform raw fixed-count sampling, bin-seeded mutations with one to three Li/vacancy swaps, and the low-bin mutation baseline. Mutation seeds come from the requested training bin; available information and attempt/oracle budgets are identified. Rejected filtered draws, duplicates and failed requests remain in accounting. Requested and measured bins, physical T, policy, teacher, schema, fit seed and sampling seed stay separate. We report confusion/hits, exact count, uniqueness, novelty to training and prior history, and distinct novel in-bin yield per all attempts. The original fixed q_{20} yield is compared only within requested-low attempts, not pooled with middle/high requests; teacher agreement is not DFT truth.

4 Temperature-dependent static acquisition

The outer policy decides how often to request each bin. Physical temperature T in kelvin does not alter the latent normal prior or introduce a decoder softmax temperature. Confirmed reactor targets motivate $T \in \{393.15, 433.15, 473.15, 493.15\}$ K; no calibrated pressure dependence or hydrothermal occupation population is assumed.

For a complete list of symmetry representatives at one composition, let i index a representative, g_i its number of distinct raw occupancy patterns, and E_{ref} a common reference energy. Define

$$a_i(T) = \exp\left[-\frac{E_i - E_{\mathrm{ref}}}{k_B T}\right], \qquad w_i = g_i a_i, \qquad W_b(T) = \frac{\sum_{i:b_i=b} w_i}{\sum_i w_i}. \tag{4}$$

Here $k_B = 8.617333262145 \times 10^{-5}$ eV/K; E_i, E_{ref} are total eV per same-composition cell. The factors a_i, w_i and bin fraction W_b are dimensionless. Stable logarithmic normalization avoids underflow. A common energy offset cancels; distorted gaps do not. Multiplicity counts physical occupation masks, not permutations of identical atom labels.

For a prespecified independent uniform *raw-mask* reference, repeated orbit identities are collapsed for energy calls but retain their raw sampling frequency f_i. Then $w_i = f_i a_i$; multiplying by g_i again would count degeneracy twice. The entire prespecified pool is scored, including exact-geometry cache coverage; using only convenient cache hits would bias the reference. Proposal-biased training or CVAE histograms are not a density of states.

The exhaustive Hu 12-site reference tests the measure: its 87 representatives cover every Li count, and orbit multiplicities sum to $\binom{12}{n}$ at each count. At $n = 6$, 17 representatives cover all 924 raw masks. Direct enumeration checks the numerical weighted sum. This small reference is not evidence for large-data generation or size transfer. For the 48-site reference, finite pool size, missing rare states and concentration remain limitations.

4.1 Concentration, mixtures, and exploration

With $Z = \sum_i w_i$ and $p_i = w_i/Z$, representative effective sample size (ESS) is $1/\sum_i p_i^2$. It describes concentration over representatives. A distinct quantity,

$$\mathrm{ESS}_{\mathrm{particle}} = \frac{Z^2}{\sum_i m_i a_i^2}, \tag{5}$$

uses $m_i = f_i$ for sampled raw draws, or $m_i = g_i$ for complete raw microstates. For equal energies it equals the number of raw draws/microstates. It is not a confidence interval. We also inspect the maximum weight and nested-prefix pool-size sensitivity; correlated prefixes are not independent repeats.

Both pure weights and an exploration policy are saved:

$$P_b(T) = (1 - \epsilon) W_b(T) + \frac{\epsilon}{K}, \qquad \epsilon = 0.1, \tag{6}$$

where K is the number of supported bins. The floor is a chosen acquisition setting, not a thermodynamic constant. Chosen temperature priorities $\alpha_t \geq 0$, $\sum_t \alpha_t = 1$, define mixtures. Largest-remainder rounding allocates exactly the declared attempt budget. Per-temperature and joint-mixture budgets are distinguished in the plan and every attempt record.

4.2 Why units and measured outcomes matter

At Li_{24}, $N_a = 168$. An illustrative 1 meV/atom gap is 0.168 eV/cell. At 433.15 K, $k_B T \simeq 0.03733$ eV and its equal-multiplicity relative weight is approximately 0.0111. Incorrectly using 0.001 eV gives approximately 0.974. Large-cell static weights can legitimately concentrate on very few reference samples; an exploration floor does not repair their physical accuracy.

Finally, teacher evaluations determine the generated masks' *actual* bins. Correct request totals do not establish conditional control or correct within-bin sampling. Because the current labels are deletion energies, this is *temperature-dependent acquisition under a static MLIP surrogate*. It is not a relaxed-basin free-energy calculation or measured configurational entropy.

5 Completed generation and acquisition measurements

Three new multibin fits completed with seeds 20261106–08, 3,476–4,187 optimizer updates, and 27.7–30.8 s fit wall time on an A100 (Adam, rate 10^{-3}, batch 512, 20-epoch warmup to $\beta = 0.1$). This excludes acquisition and scoring. The frozen balanced benchmark contains 41,472 attempts (256 per method/count/request/replicate) and 40,662 distinct geometries. All masks were valid and exact-count, with complete UMA and MACE scalar labels and no request failures. Novelty is absent from both 40,157 training keys and 188,919 frozen prior-history keys. CVAE fit seeds and baseline random replicates are distinct.

Method	UMA useful yield (%)	MACE useful yield (%)
Uniform raw masks	33.00 ± 0.65	33.38 ± 0.26
AR, composition only	31.73 ± 0.77	31.94 ± 0.85
AR, binary energy	31.92 ± 1.01	31.03 ± 1.11
AR, three energy bins	31.00 ± 0.81	31.24 ± 1.22
Requested-bin seed mutation	58.39 ± 0.61	57.54 ± 0.16
Low-bin seed mutation	33.93 ± 1.05	34.40 ± 0.82

Distinct novel in-requested-bin orbits / all attempts, macro-averaged equally over three counts and three requested bins, then three design seeds. ± is descriptive seed standard deviation, not a confidence interval. MACE uses its own train-derived energy boundaries. Mutation controls use the same round-0 training information, with no fitted model.

The multibin extension did not improve this balanced objective. UMA requested-bin hit rates for low/middle/high were 35.59/54.43/8.42%, versus 16.02/63.32/19.70% for uniform sampling and 49.18/71.48/54.60% for bin-seeded mutation. The generator favors low energies but does not reliably distinguish all three requests. This does not isolate why; a controlled training-weight ablation and condition-sensitivity checks remain assigned. On requested-low attempts, the separate original-q_{20} UMA yield was 32.99% for multibin, 33.94% binary, 17.49% uniform and 51.65% bin mutation. These use the new history/budget and are not the historical values on page 5.

5.1 Executed acquisition demonstration

The three fits executed all 90 prespecified per-temperature and mixture schedules: 11,520 attempts, 11,080 geometries, no failed requests, complete two-teacher scoring. Under pure per-temperature requests, 100% were asked to be low, but measured low fractions were only 36.48% UMA and 40.26% MACE. The floor requested 93.75/3.125/3.125% after rounding; measured low fractions were 35.79% and 39.24%. Correct budget execution did not overcome weak conditional control. The independent reference is fully scored: 24,576 raw draws collapse to 24,575 geometries. At 433.15 K its particle ESS is 4.629/2.271/5.480 for $n = 12/24/36$, with maximum representative mass 0.361/0.654/0.299. Across the temperature grid, low-bin mass exceeds 99.986%, so every rounded 128-attempt static request is 128/0/0 and the exploration request is 120/4/4. Temperature changes weights/ESS, but not these coarse integer bin allocations. Nested 512-to-8,192-draw prefixes find minimum-energy decreases of 0.0101/0.1758/0.2459 eV/cell. Thus stable bin fractions do not establish adequate low-tail coverage or equilibrium accuracy.

The new campaign used 127,481 logical single-point energy/force/stress calls, including 24 interface-pilot calls, and 12.401 allocated GPU-device hours (3.100 four-GPU-equivalent node-hours). Both allocations are released. These costs include startup, failures, analysis and idle time; inherited historical labels are additional prior costs, not free end-to-end inputs. Neural-backbone forward calls were not separately instrumented.

Sources: new benchmark `summary.csv`, `summary_by_request.csv`, `fixed_q20_low_request.csv` and `receipt.json`; reference diagnostics `prefix_metrics.csv` and `allocation_fractions.csv`; demo `receipt.json` and `allocation_realization.csv`; `cost_receipt.json`.

6 Historical context, interpretation, and remaining checks

The completed November 6 binary experiment used three CVAE fit seeds and three random replicates for sampler controls, three Li counts, and 512 attempts per group. Its original teacher-specific low-tail thresholds and novelty boundary differ from a later assessment and are preserved as historical evidence. Yield counts distinct novel below-threshold orbits per all attempts, including repeats:

Historical method	UMA yield (%)	MACE yield (%)
Uniform fixed count	18.99	19.75
AR CVAE, composition only	31.88	35.76
AR CVAE, binary energy	35.57	40.06
Additional adaptive AR, $\beta = 0.1$	33.81	38.91
Strongest mutation seed pool	52.65	54.08

The binary CVAE beat random sampling but not strong mutation. Additional adaptation coupled data, thresholds and class weights, so its decline does not isolate a cause. A separate geometry CVAE reduced the matched optimizer's median steps from 40 to 24, but a deterministic regressor also reached 24; no CVAE-specific geometry advantage was established. These are secondary older results, not outcomes of the new multibin extension.

6.1 How far can teacher rankings be trusted?

A new recomputation of the existing exact-geometry audit keeps DFT source, phase, composition and cohort separate. Across seven historical O3 rattled pools (524 frames), UMA median Spearman correlation is 0.97366, low-quintile precision 96.30%, and pair-gap root-mean-square error (RMSE) 2.047 meV/atom. Across ten informative Hu O3 optimized pools (104 frames), these medians are 0.73177, 41.67%, and 4.077 meV/atom. Pool sizes/ties and all three models are tabulated separately. Strongly displaced frames can give more favorable rank correlations than nearly degenerate optimized structures.

Hu's optimized $Li_{10}Co_{12}O_{24}$ pool contains five structures. UMA picks `structure_4159` first although DFT ranks it fifth; its selection regret is 7.9711 meV/atom relative to the best DFT member of that complete pool. Matching all five identities is trivial top-five overlap here, not recall from a larger pool. Historical/Hu DFT settings differ, and archived electronic/spin convergence is not recertified. This audit used FP64; the new deletion benchmark uses its separately pinned FP32 protocol. Good ranks do not establish accurate gaps, forces, phonons, or thermal weights.

6.2 Novelty, limits, and the DFT boundary

The contribution is a reproducible test of composition/energy-range control, strong baselines, symmetry-aware novelty/cost, and a measured acquisition layer. It does not require the CVAE to win. It does not establish global minima, DFT accuracy of new 48-site structures, physical entropy, full phase stability, reactor optima, downstream MLIP improvement, or size/material transfer. A later same-potential thermal pilot is kept separate from this class report. Physical accuracy requires consistent matched DFT references; no new DFT is submitted or scheduled in this work.

Independent team contributions remain: Beihao reproduces a multibin fit on the class GPU, explains the implemented model, and supplies learning curves and one controlled ablation. Varad independently recomputes historical/new metrics, multiplicities, allocation and costs, and prepares seed-resolved figures. Ricardo's physical motivation, data/site/symmetry provenance, ranking audit and interpretation are supplied here. Individual checks are targeted for November 7, integration/table freeze November 8, and final verification November 9 before 11:59 p.m. Pacific. Prepared infrastructure does not count as collaborators' completed checks; no Gradescope submission is automated.

References

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